Forecasting and Data Analytics

MGT 207: Production and Operations Management

Session 4: Time Series, Decomposition (Prophet), Causal AI & Simulation Strategy

Forecasting Methods Defined

Expert Forecast (Subjective)

Qualitative Method

A method based on domain judgment and executive consensus.

- **Key Strength:** Incorporates specialized market intuition or regime changes difficult to codify mathematically.

Time-Series Forecasting

Quantitative Method

Built on the premise that historical demand structure predicts future trajectory.

- **Data Usage:** Extracts level, trend, and seasonal cycles across discrete time periods.

Regression (Causal)

Statistical Method

Models cause-and-effect relationships between dependent (Y) and independent (X) variables.

- **Approach:** Links demand to quantifiable levers (price, promotions, ad spend, weather).

AI / Foundation Models

Advanced Quantitative

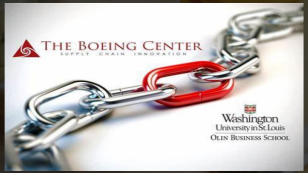
Combines machine learning (XGBoost), automated decomposition (Prophet), and time-series foundation models.

- **Scale:** Captures non-linear interactions across thousands of SKUs and exogenous signals.

ANHEUSER-BUSCH

Example

Demand Forecasting with Machine Learning



LIFT

LOGISTICS INVENTORY AND FORECASTING TEAM

A/B “Why Forecasting”

A/B “Why”

Forecast Accuracy directly relates to the amount of Out-of-Stocks (OOS) at AB’s WODs (Wholesale Operation Divisions)

Total Benefit

-4% OOS for each +1% Accuracy

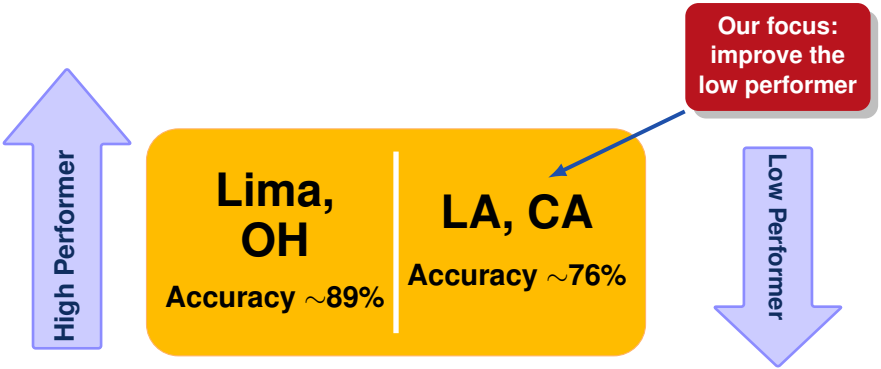
High End Benefit

-11% OOS for each +1% Accuracy

Background

Background

A/B has historically had mixed success with forecasting based on WOD location:



Time-Dependent & Exogenous Factors

Focusing on internal time-series patterns and external causal drivers:

- **Sales History**
  - Previous week, biweekly, monthly, 6-month, yearly
- **Holidays**
  - Memorial Day, Labor Day, Independence Day (2 weeks before and after)
- **Seasonality**
  - Yearly, quarterly, monthly cycles
- **Promotion**
  - Average discount rates and promotional windows
- **Weather**
  - Average temperature and precipitation anomalies
- **Brand Characteristics**
  - Frontline price, wholesale price to retail

Time Series-Based Model & Today's Two Turnarounds

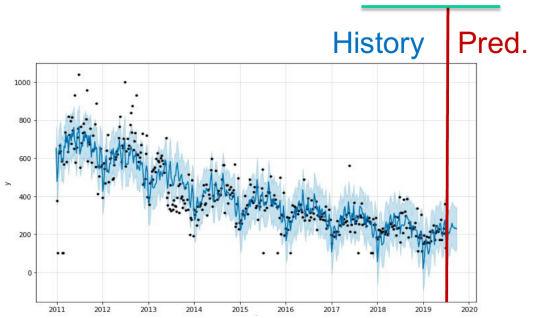
Use LA Data

Prediction accuracy:

~93%

Utilized Data:

- Past sales history (declining trend)
- Strong summer seasonality
- Weather & holiday promos



Two Parallel Legacy Brand Turnarounds Today: **(1) Hour 1 — Anheuser-Busch LA Legacy Lager:** secular downward trend (-20k cases/qtr) + strong Q3 summer peaks & Q1 winter troughs. **(2) Hour 2 — Harvard Simulation (Kelsey-White "Blue" Detergent):** declining national pod/liquid share requiring regional trend, causal pricing, and Newsvendor production strategy.

The Nuts and Bolts of

TIME SERIES FORECASTING

Building Intuition, Baselines, and the Lag vs. Noise Trade-Off

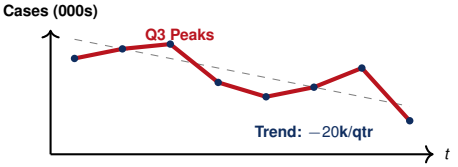
Timeline Notation ($A_{t-1} \rightarrow F_t$) & Running Case: A/B LA Lager

Core Forecasting Notation

- $\mathbf{A_{t-1}}$: Actual demand observed in period $t-1$.
- $\mathbf{F_t}$: Forecast made in $t-1$ for upcoming period t.

The Golden Timeline Rule

At the end of period $t-1$, observe actual A_{t-1} and lock in production forecast F_t for period t.



A/B Los Angeles WOD — Legacy Lager (000s Cases)

Period (t)	Quarter	Actual (A_t)	Underlying Market Structure
1	Y1 Q1 (Win)	**450**	Baseline 480 − 30 (Winter trough)
2	Y1 Q2 (Spr)	**470**	Baseline 460 + 10 (Spring warm-up)
3	Y1 Q3 (Sum)	**480**	Baseline 440 + 40 (Summer peak)
4	Y1 Q4 (Fall)	**400**	Baseline 420 − 20 (Fall cooldown)
5	Y2 Q1 (Win)	**370**	Baseline 400 − 30 (Winter trough)
6	Y2 Q2 (Spr)	**390**	Baseline 380 + 10 (Spring warm-up)
7	Y2 Q3 (Sum)	**430**	360 + 40 + **30** (96°F + Labor Day promo)
8	Y2 Q4 (Fall)	**320**	Baseline 340 − 20 (Fall cooldown)
9	**Y3 Q1 (Win)**	**?**	**True Baseline** = 320 − 30 = 290

Structural Equation

$\text{Baseline}_t = 500 - 20t$;
$S_{Q1} = -30, S_{Q2} = +10, S_{Q3} = +40, S_{Q4} = -20.$

Window Averaging: Naïve, Simple Average & 4-Quarter MA (MA_4)

t	Qtr	A_t	Naïve (A_{t-1})	MA_4	MA_4 Calculation
1	Y1 Q1	450	–	–	Need 4 quarters
2	Y1 Q2	470	450.0	–	Need 4 quarters
3	Y1 Q3	480	470.0	–	Need 4 quarters
4	Y1 Q4	400	480.0	–	Need 4 quarters
5	Y2 Q1	370	400.0	**450.0**	$(450+470+480+400)/4$
6	Y2 Q2	390	370.0	**430.0**	$(470+480+400+370)/4$
7	Y2 Q3	430	390.0	**410.0**	$(480+400+370+390)/4$
8	Y2 Q4	320	**430.0**	**397.5**	$(400+370+390+430)/4$
9	**Y3 Q1**	**?**	**320.0**	**377.5**	$(370+390+430+320)/4$

Formulas: Naïve: $F_t = A_{t-1}$ — Moving Avg:

$$F_t = \frac{1}{n}\sum_{i=1}^{n} A_{t-i}$$

1. Why Naïve Fails ($t = 8$ Whipsaw)

After summer peak $A_7 = 430$, Naïve predicts $F_8 = 430$ for Fall. Actual demand drops to $A_8 = 320$ $\Longrightarrow$ **110,000 cases of Fall warehouse overstock!**

2. Why MA_4 Fails (Trend Lag)

Averaging 4 quarters ($n = 4$) cancels out seasonal swings ($\sum S_Q = 0$), **but severely lags the -20k/qtr decline**:

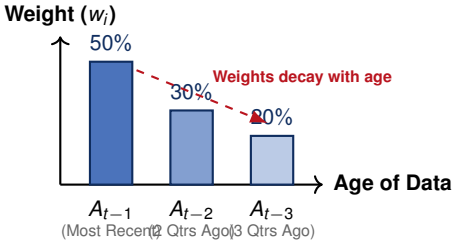
- Every MA_4 forecast ($t = 5..8$) is mostly **above actual demand**.
- For $t = 9$, MA_4 predicts **377.5**k vs. true winter baseline of **290**k ($+87.5$k overprediction!).

Weighted Moving Average (WMA) & Exponential Smoothing

1. Weighted Moving Average (WMA)

Assigns **more weight to recent data** and less weight to older data ($\sum w_i = 1.0$):

$$F_t = 0.50\,A_{t-1} + 0.30\,A_{t-2} + 0.20\,A_{t-3}$$



2. Simple Exponential Smoothing (SES)

Uses **just one weight** $\alpha \in [0, 1]$ to blend last period's actual demand with last period's forecast:

$$\mathbf{F_t = \alpha A_{t-1} + (1 - \alpha) F_{t-1}}$$

or equivalently: $\mathbf{F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1})}$

Why Managers Prefer Exponential Smoothing

- **Error Correction:** Adjusts F_{t-1} by a fraction α of last period's error ($A_{t-1} - F_{t-1}$).
- **One Tuning Knob (α):** Small α (0.2) = stable baseline; Large α (0.8) = fast reaction.
- **Minimal Storage:** Requires only **two numbers** (A_{t-1} and F_{t-1}) per SKU!

Exponential Smoothing in Action ($\alpha = 0.2$ on A/B LA Demand)

Parameters: $\alpha = 0.2$, **Initial Baseline** $F_1 = 480.0$

t	Qtr	A_t	SES (F_t)	**Recursive Calculation:** $0.2A_{t-1} + 0.8F_{t-1}$
1	Y1 Q1	450	480.0	Given initial level baseline ($F_1 = 480$)
2	Y1 Q2	470	**474.0**	$0.2(450) + 0.8(480.0) = 90.0 + 384.0 = \mathbf{474.0}$
3	Y1 Q3	480	473.2	$0.2(470) + 0.8(474.0) = 94.0 + 379.2 = 473.2$
4	Y1 Q4	400	474.6	$0.2(480) + 0.8(473.2) = 96.0 + 378.6 = 474.6$
5	Y2 Q1	370	459.6	$0.2(400) + 0.8(474.6) = 80.0 + 379.6 = 459.6$
6	Y2 Q2	390	441.7	$0.2(370) + 0.8(459.6) = 74.0 + 367.7 = 441.7$
7	Y2 Q3	430	431.4	$0.2(390) + 0.8(441.7) = 78.0 + 353.4 = 431.4$
8	Y2 Q4	320	431.1	$0.2(430) + 0.8(431.4) = 86.0 + 345.1 = 431.1$
9	**Y3 Q1**	**?**	**408.9**	$\mathbf{0.2(320) + 0.8(431.1) = 64.0 + 344.9 = 408.9}$

Initialization Dilemma: 2 Options

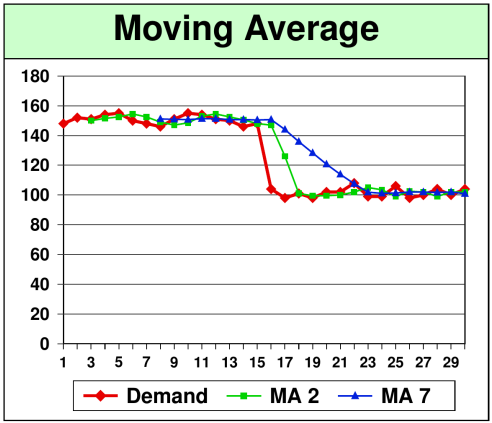
1. **Naïve Start:** Set $F_2 = A_1 = 450$, then smooth from $t = 3$ onward.
2. **Baseline Start (Used Here):** Set $F_1 = 480$ from prior-year executive baseline.

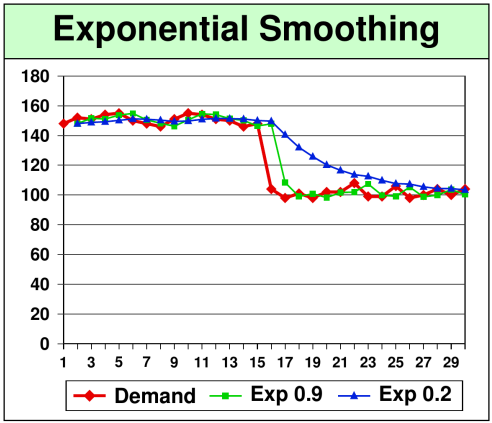
Why SES Fails in LA's Declining Market

Because $\alpha = 0.2$ places **80% weight on past high history**, F_t stays stubbornly above actual sales in 7 of 8 quarters!

- Mean Forecast Error ($\text{MFE}_{1..8}$) $= -\mathbf{44.5}$k cases/qtr (chronic overstocking).
- Predicts $F_9 = 408.9$k vs. 290k winter baseline!

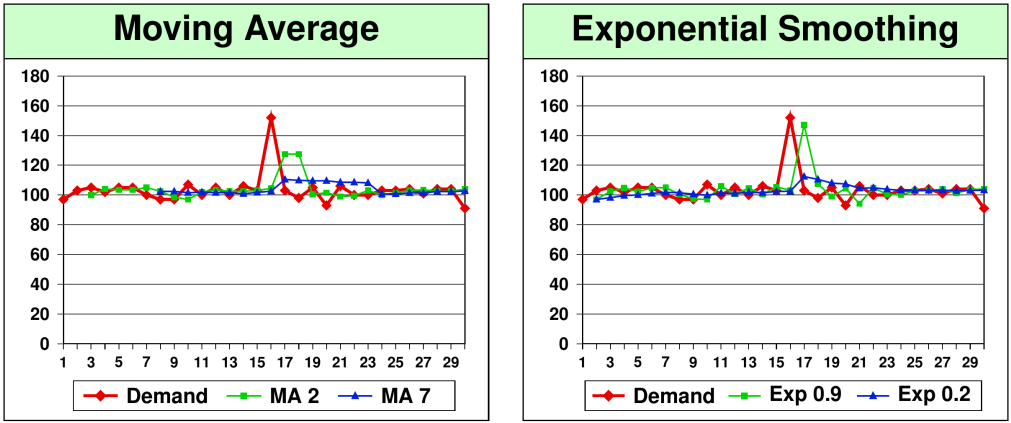
Forecast Parameter Influences Responsiveness to Shift in Mean





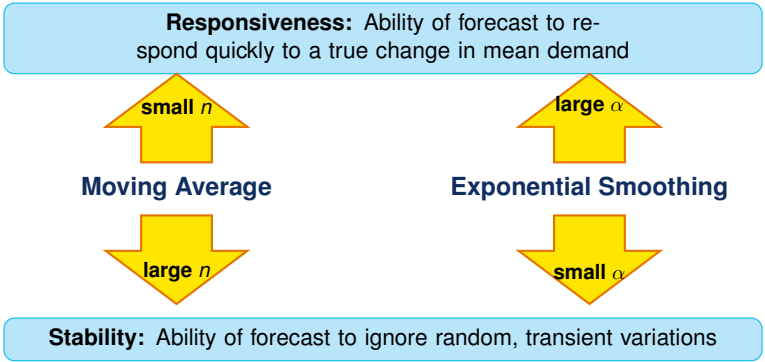
- **Moving Average:** Small n (MA 2) tracks the downward drop quickly; large n (MA 7) lags above it.
- **Exponential Smoothing:** Large α (Exp 0.9) adjusts immediately; small α (Exp 0.2) lags and overpredicts.

Forecast Parameter Influences Stability to Random Fluctuations



- **Moving Average:** Large *n* (MA 7) filters out noise; small *n* (MA 2) overreacts to spikes.
- **Exponential Smoothing:** Small α (Exp 0.2) maintains baseline; large α (Exp 0.9) chases noise.

Responsiveness vs. Stability Trade-Off

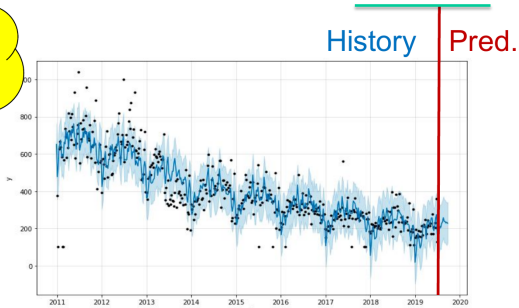


Preview of Hour 2 Simulation (Kelsey-White's "Blue" Detergent)

In Hour 2, you will manage 4 years of turnaround decisions for **"Blue" laundry detergent** across 4 U.S. regions. Clicking the simulator's *"Fill in previous year's value"* button is a pure **Naïve Forecast** ($\alpha = 1, n = 1$)—it chases last year's noise while ignoring regional trends!

Time Series with Trends & Seasonality

What would happen if we applied Exponential Smoothing to the A/B demand pattern?



Simple Exponential Smoothing: Lag in Trends

Systematic Forecast Error

In a declining market, simple exponential smoothing **overpredicts** demand because it lags behind the trend. In a growing market, it systematically **underpredicts** demand.



Exponential Smoothing with Trend Adjustment (DES / Holt's Method)

Governing Equations ($\alpha, \beta \in [0, 1]$)

1. Smoothed Level (F_t):

$$F_t = \alpha A_{t-1} + (1 - \alpha)(F_{t-1} + T_{t-1})$$

2. Smoothed Trend (T_t):

$$T_t = \beta(F_t - F_{t-1}) + (1 - \beta)T_{t-1}$$

3. Forecast Including Trend (FIT_t):

$$\mathbf{FIT_t = F_t + T_t}$$

Step 1: Update Smoothed Level (F_t)
Blends latest actual A_{t-1} with prior trend-adjusted forecast ($F_{t-1} + T_{t-1}$).

Step 2: Update Smoothed Slope (T_t)
Blends latest level change ($F_t - F_{t-1}$) with prior trend T_{t-1} using weight β.

Step 3: Combine into FIT_t
$\mathbf{FIT_t = F_t + T_t}$ eliminates systematic trend lag!

DES in Action on A/B LA Demand ($\alpha = 0.2, \beta = 0.4$)

Initial: $F_1 = 480.0, T_1 = -20.0 \implies FIT_1 = 460.0$

t	Quarter	A_t	Level (F_t)	Trend (T_t)	$\mathbf{FIT_t = F_t + T_t}$
1	Y1 Q1 (Win)	450	480.0	−20.0	**460.0**
2	Y1 Q2 (Spr)	470	**458.0**	**−20.8**	**437.2**
3	Y1 Q3 (Sum)	480	443.8	−18.2	**425.6**
4	Y1 Q4 (Fall)	400	436.5	−13.8	**422.7**
5	Y2 Q1 (Win)	370	418.1	−15.6	**402.5**
6	Y2 Q2 (Spr)	390	396.0	−18.2	**377.8**
7	Y2 Q3 (Sum)	430	380.2	−17.2	**363.0**
8	Y2 Q4 (Fall)	320	376.4	−11.9	**364.5**
9	**Y3 Q1 (Win)**	**?**	**355.6**	**−15.5**	**340.1**

Worked Calculation: Period $t = 2$

- **Step 1 (Level):**
  $F_2 = 0.2(450) + 0.8(480.0 - 20.0) =$
  $90.0 + 368.0 =$ **458.0**
- **Step 2 (Trend):**
  $T_2 = 0.4(458.0 - 480.0) + 0.6(-20.0) =$
  $-8.8 - 12.0 =$ **−20.8**
- **Step 3 (Forecast):**
  $FIT_2 = F_2 + T_2 = 458.0 - 20.8 =$ **437.2**

What DES Fixes — and What It Misses

DES tracks the −20k/qtr slope ($FIT_9 = 340.1$ vs. SES 408.9), cutting bias to near zero! **But** it still misses +40k summer peaks and −30k winter troughs.

Capturing the Wave: Seasonal Decomposition (Holt-Winters Intuition)

Additive Decomposition Architecture

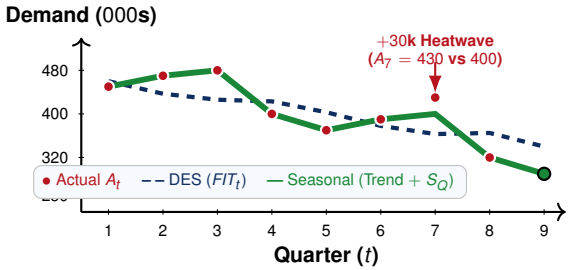
$$\text{Forecast}_t = \text{Level}_t + \text{Trend}_t + \text{Seasonal}_Q$$

- **Trend Baseline:** $\text{Level}_t + \text{Trend}_t = 500 - 20t$
- **Quarterly Seasonal Indices (S_Q):**
  - Q1 (Win) $= -\mathbf{30}$k, Q2 (Spr) $= +\mathbf{10}$k
  - Q3 (Sum) $= +\mathbf{40}$k, Q4 (Fall) $= -\mathbf{20}$k

Predicting Y3 Q1 ($t = 9$) with Seasonality

Trend Baseline$_9 = 500 - 20(9) = 320$k.
Add Q1 Winter Index ($S_{Q1} = -30$k):

$$\mathbf{F_9} = \mathbf{320} - \mathbf{30} = \mathbf{290}.\mathbf{0}\text{k cases!}$$



Modern Automated Decomposition: Meta's Prophet (Prophet_LA.ipynb)

The Engine Behind A/B's 93% LA Forecast (slide09_chart.png & slide26_chart.png)

$$\mathbf{y(t)} = \mathbf{g(t)}_{\text{Piecewise Trend}} + \mathbf{s(t)}_{\text{Fourier Seasonality}} + \mathbf{h(t)}_{\text{Holiday Windows}} + \epsilon_{\mathbf{t}}$$

Why Operations Teams Use Prophet

- **Automatic Changepoints ($g(t)$):** Detects when a brand's growth slope bends downward without manual re-initialization.
- **Multi-Period Seasonality ($s(t)$):** Simultaneously fits weekly retail cycles and annual summer/winter waves.
- **Holiday Windows ($h(t)$):** Explicitly models ± 2 week surges around July 4th, Memorial Day, and Labor Day.

Industry Standard Baseline

- Bridges classic Holt-Winters smoothing and modern machine learning.
- Robust to missing data, outliers, and SKU-level noise.
- **What's Still Missing?** Dynamic causal levers (price changes, promo discounts, competitor actions, heatwaves) $\Longrightarrow$ **Module 4: Causal Models!**

Simulation Connection: Regional Trend Forecasting for "Blue" Detergent

What Is Today's Harvard Simulation About? (Executive Context)

You are the Turnaround Brand Manager for Kelsey-White's flagship laundry detergent, **"Blue,"** competing in a 4-brand market (*Blue* [$7], discount *Store* [$6], mid-tier *Fresh* [$8], and premium *Turbo* [$10]). Each year (Years 5–8), you set **Formulation** (*Powder, Liquid, Pods*), **Channel Price**, **Media/Trade Spend**, and **Regional Demand Forecasts** across 4 U.S. regions to guide your national **Production Order**.

K-W Historical Demand: West Region (Years 1–4)

Year	West Demand (M)	YoY Growth	Trajectory
Year 1	11.0M	–	Baseline
Year 2	11.1M	+0.9%	Flat start
Year 3	12.5M	+12.6%	Strong expansion
Year 4	13.4M	+7.2%	Continued growth
Y5 (4-Yr MA)	**12.0**M	Severe lag (−2.4M underprediction!)	
Y5 (Naïve)	**13.4**M	Ignores upward regional momentum	
Y5 (Trend-Adj)	**14.4**M	$13.4\text{M} \times (1 + 0.072) = \mathbf{14.36}\text{M}$	

Step 1 of the Simulation Playbook

- Each of the 4 regions (*North, South, East, West*) has its own secular trend.
- Clicking *"Fill in previous year's value"* (13.4M) or using MA_4 (12.0M) in a growing region like the West **underforecasts** and starves sales!
- **Rule:** Project the **Trend-Adjusted Baseline** first (14.4M), then adjust for your Year 5 causal levers (Price, Pods, Media).

Why Run Regression on A/B Data? Passive vs. Active Forecasting

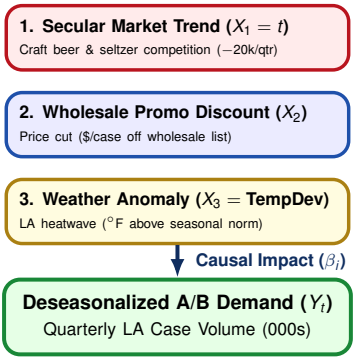
Why Time Series Alone Fails in a Declining Market

- **Time Series Is Passive:** Extrapolating calendar time (t) only tells A/B that LA demand is falling by -20k cases/quarter.
- **Missed the Y2 Q3 Surge ($t = 7$):** Pure seasonal decomposition predicted 400k cases, but actuals hit **430**k (30k stockout!).
- **Managers Pull Levers:** To fight a declining market, A/B runs **Wholesale Price Promos** and reacts to **LA Heatwaves**.

Multiple Linear Regression Model

$$\mathbf{Y_t} = \beta_0 + \beta_1\,\mathrm{Time_t} + \beta_2\,\mathrm{Promo_t} + \beta_3\,\mathrm{TempDev_t} + \epsilon_t$$

Separates **secular decline ($\beta_1 < 0$)** from **promo lift ($\beta_2 > 0$)** and **weather shocks ($\beta_3 > 0$)**!



Reading A/B LA's Regression Output Like an Executive

Dependent Variable: Deseasonalized A/B LA Demand (000s cases)

Regression Statistics		ANOVA	df	F	Significance F
Multiple R	0.9920	Regression	3	82.45	**0.0004**
R Square	**0.9841**	Residual	4		
Adjusted R Square	0.9722	Total	7		
Observations	8 Quarters				

Variable	Coeff (β_i)	Std Error	t Stat	P-value	Managerial Meaning
Intercept (β_0)	**500.0**	4.12	121.36	**0.0001**	Starting LA baseline (500k)
Time Trend (t)	$-$**20.0**	0.78	-25.64	**0.0001**	Losing **20**k cases/quarter
Promo ($/case off)	$+$**10.0**	1.45	6.90	**0.0023**	$+$**10**k cases per $1 price cut
TempDev (°F > norm)	$+$**1.5**	0.28	5.36	**0.0058**	$+$**1.5**k cases per $+1$°F heat

3-Point Executive Checklist

1. **Model Fit ($R^2 = 98.4\%$):** Trend, Promo, and Weather explain 98.4% of deseasonalized variance.
2. **Signal vs. Noise ($p < 0.05$):** All 3 drivers have $p < 0.01$ (statistically significant, not random noise).
3. **Actionable Levers (β_i):** A \$2.00/case promo discount ($+$20k) exactly offsets one quarter of secular decline ($-$20k)!

Multivariate Demand Model: Explaining A/B LA's Declining Curve

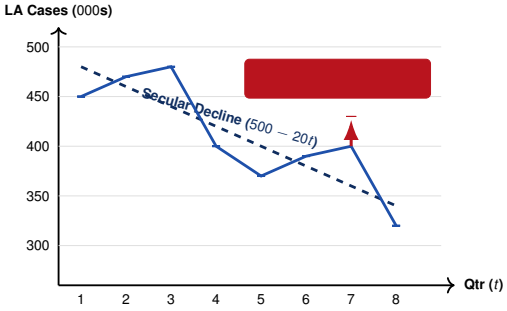
Full Empirical Model (Trend + Seasonality + Causal)

$$\widehat{\text{Cases}}_t = (\mathbf{500} - \mathbf{20}\,t + \mathbf{S_Q}) + \mathbf{10.0}\,\text{Promo}_t + \mathbf{1.5}\,\text{TempDev}_t$$

Solving the Y2 Q3 ($t = 7$) 430k Mystery

Why did Y2 Q3 jump to 430k when the declining trend + summer wave predicted 400k?

- **Seasonal Baseline ($t = 7, S_{Q3} = +40$):** $500 - 20(7) + 40 = \mathbf{400}$k
- **Labor Day Promo ($1.50/case off):** $+10.0 \times 1.50 = +\mathbf{15}$k
- **LA Heatwave ($+10°$F above norm):** $+1.5 \times 10 = +\mathbf{15}$k
- **Causal Forecast:** $400 + 15 + 15 = \mathbf{430}$k cases (Zero stockout!)



Actionable Forecasting & "What-If" Scenario Testing at A/B LA

Planning Year 3 Q1 ($t = 9$, Super Bowl Quarter in LA)

- **Passive Baseline (Promo** $= \$0$, **TempDev** $= 0°$**F):** $\widehat{\text{Cases}}_9 = 500 - 20(9) - 30_{\text{Win}} =$ **290**k cases.
- **Active Super Bowl Scenario (**$\$2.00/$**case Promo +** $4°$**F Warmer Winter):**

$$\widehat{\text{Cases}}_{\mathbf{9}} = \mathbf{290}\text{k (Baseline)} + \mathbf{10.0(2.00)}_{\text{Promo}} + \mathbf{1.5(4)}_{\text{Weather}} = \mathbf{290} + \mathbf{20} + \mathbf{6} = \mathbf{316}\text{k cases (+26k lift!)}$$

Why Regression Matters in a Turnaround

- **Quantifies Trade-Offs:** Tells executives exactly how much price promotion is required to stem a -20k/qtr market decline.
- **Prevents Promo Stockouts:** If marketing runs a $2.00 Super Bowl promo but operations orders the 290k passive forecast, LA runs out of beer by $26,000$ cases!

Direct Link to "Blue" Detergent

- Just like Budweiser, **"Blue"** is fighting a turnaround battle.
- When you cut price (e.g., $\$7.00 \rightarrow \$5.50$) or launch **Pods** in Year 5, you are pulling causal regression levers—your forecast *must* include that lift!

Simulation Connection: The Causal Levers of "Blue" Detergent

Implicit Multiple Regression Inside K-W Vision

$$\text{Demand}_{\mathbf{r},\mathbf{t}} = \text{Trend}_{\mathbf{r},\mathbf{t}} + \beta_{\text{Price}}\text{Price}_{\mathbf{t}} + \beta_{\text{Form}}\text{Formulation}_{\mathbf{t}} + \beta_{\text{Pos}}\text{Positioning}_{\mathbf{t}} + \beta_{\text{Media}}\text{Media}_{\mathbf{r},\mathbf{t}}$$

1. Pricing & Formulation Levers

- **Channel Price ($\beta_{\text{Price}} < 0$):** Cutting price from \$7.00 to \$5.50 drives strong volume elasticity (vs. *Store* \$6, *Fresh* \$8, *Turbo* \$10), but watch unit margin.
- **Formulation (β_{Form}):** Upgrading from **Powder** to **Liquid** ($+7\%$ cost) or **Pods** ($+15\%$ cost) unlocks high growth among younger cohorts.

2. Positioning & Demographic Fit

- **Product Positioning:** Align *Odor Elimination* vs. *Cold Water Wash* with regional preferences.
- **Media & Trade Spend:** Match **Print vs. Digital Media** to regional age cohorts (younger < 45 regions respond strongly to Digital + Pods).

The Executive Analytics Trap: Spurious Correlations & Noisy Dashboards

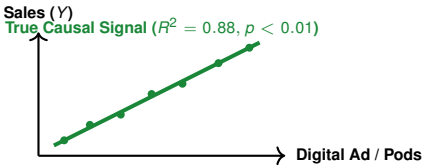
Not Every Dashboard Widget Is Causal!

BI dashboards (like **K-W Vision**) display many charts. Before acting on a metric, verify:

1. **Slope & Fit (R^2):** Tight linear slope vs. flat scatter cloud ($R^2 \approx 0$)?
2. **Significance ($p < 0.05$):** True signal or random sample noise?
3. **Economic Logic:** Plausible mechanism linking X to purchases?

Simulation Audit Challenge

In K-W Vision, inspect the **Social Media Sentiment** tab: does sentiment correlate with unit sales ($R^2 > 0$), or is it a **zero-correlation distraction**? Test before you trust!



Measuring Forecast Errors

Auditing AI & Statistical Models, Detecting Bias, and Sizing Inventory Buffers

The 4 Essential Error Metrics: Formulas & Managerial Meaning

Forecast Error in Period t: $\mathbf{e_t = A_t - F_t}$ ($e_t > 0 \implies$ Underforecast / Stockout; $e_t < 0 \implies$ Overforecast / Excess Inventory)

1. Mean Forecast Error (MFE) — Bias

$$\text{MFE} = \frac{1}{n}\sum_{t=1}^{n}(\mathbf{A_t - F_t})$$

- Measures **directional bias** ($+$ and $-$ errors cancel). Target MFE ≈ 0.

2. Mean Absolute Deviation (MAD) — Magnitude

$$\text{MAD} = \frac{1}{n}\sum_{t=1}^{n}|\mathbf{A_t - F_t}|$$

- Average miss in **physical units**. Directly sizes safety stock ($\sigma \approx 1.25\,\text{MAD}$).

3. Mean Absolute Percentage Error (MAPE)

$$\text{MAPE} = \frac{1}{n}\sum_{t=1}^{n}\frac{|\mathbf{A_t - F_t}|}{\mathbf{A_t}} \times \mathbf{100}\%$$

- **Scale-independent KPI:** Compares WODs of any size (Lima 11% vs. LA 24%).

4. Mean Squared Error (MSE) — Outlier Penalty

$$\text{MSE} = \frac{1}{n}\sum_{t=1}^{n}(\mathbf{A_t - F_t})^2$$

- Squares errors, heavily penalizing **catastrophic stockouts or gluts**.

Model Audit & Horse-Race on A/B LA Demand ($t = 5..8$, Year 2)

Worked Audit: DES (FIT_t) in Year 2 ($t = 5..8$)

t	A_t	FIT_t	$e_t = A_t - FIT_t$	$\lvert e_t \rvert$	$\lvert e_t \rvert / A_t$	e_t^2
5	370	402.5	-32.5	32.5	8.8%	1,056
6	390	377.8	$+12.2$	12.2	3.1%	149
7	430	363.0	$+67.0$	67.0	15.6%	4,489
8	320	364.5	-44.5	44.5	13.9%	1,980
Avg	**377.5**	**377.0**	$+$**0.6**	**39.1**	**10.4%**	**1,919**

Key Diagnostic Insight

DES eliminates directional bias (MFE $= +0.6$k vs. -63.5k for SES!), but its MAD $= 39.1$k remains high because it ignores seasonality.

A/B LA Model Leaderboard (Year 2: $t = 5..8$)

Forecasting Model	MFE (Bias)	MAD	MAPE	MSE
1. Naïve (A_{t-1})	-20.0k	50.0k	14.3%	3,950
2. 4-Qtr MA (MA_4)	-44.4k	54.4k	15.6%	3,152
3. Simple ES ($\alpha = 0.2$)	-63.5k	64.2k	18.6%	5,705
4. DES Trend (FIT_t)	$+$**0.6**k	39.1k	10.4%	1,919
5. Seasonal Prophet	$+7.5$k	**7.5**k	**1.7%**	**225**
6. Prophet + Causal	**0.0**k	**2.1**k	**0.5%**	**6**

Result: Moving from Simple ES (81.4% accuracy) to Prophet + Causal Weather/Promo achieves A/B's $>$ **93**% target!

MAD vs. MSE Decision Rule & Setting Simulation Bounds

MAD vs. MSE: Linear vs. Quadratic

- **MAD (Linear):** A 20k error is penalized 2× a 10k error. Best for communicating typical unit deviation.
- **MSE (Quadratic):** A 20k error (400) is penalized **4**× a 10k error (100).

Executive Decision Rule

To avoid occasional **catastrophic stockouts or massive warehouse gluts** (even at the cost of frequent small misses), **minimize MSE**!

Simulation Connection: Setting Min & Max Bounds

In Harvard's Simulation, you enter **Low (Min)**, **Expected**, and **High (Max)** for each region's triangular distribution:

- **Expected:** Your Trend + Causal Regression point forecast ($\hat{Y}_r$).
- **Low / High Bounds:** Use historical forecast error dispersion (MAD_r):

$$\text{Min}_\mathbf{r} \approx \hat{\mathbf{Y}}_\mathbf{r} - \mathbf{2} \times \text{MAD}_\mathbf{r}, \quad \text{Max}_\mathbf{r} \approx \hat{\mathbf{Y}}_\mathbf{r} + \mathbf{2} \times \text{MAD}_\mathbf{r}$$

- Never guess bounds randomly—ground them in your model's empirical error!

AI-Powered Forecasting: The Modern Enterprise Landscape

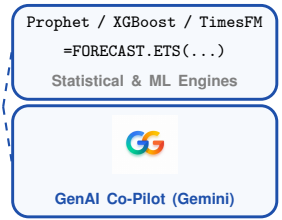
Two Distinct Pillars of AI in Forecasting

1. Predictive ML & Time-Series Foundation Models

- *Google TimesFM, Amazon Chronos, LightGBM / XGBoost.*
- Trained on billions of time-series points to merge autoregression, seasonality, and hundreds of causal variables at SKU scale.

2. Generative AI / LLMs as Analytics Co-Pilots (*Gemini*)

- LLMs do **not** "guess" numbers from text tokens; they **write and execute Python/Prophet/OLS code**, tune (α, β), and stress-test scenarios.



The Winning Enterprise Stack: Rigorous Statistical/ML Engines + GenAI Co-Pilot Execution + Managerial Domain Judgment.

Workflow 1: Using Gemini for Time-Series & Prophet Modeling

1. Data Integration

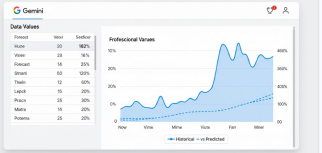
Upload historical demand spreadsheets directly (e.g., A/B WOD shipments or K-W regional histories).

2. Automated Model Horse-Race

- Writes & executes Python code to run MA_4, SES, DES, and Meta Prophet side by side.
- Grid-searches optimal smoothing parameters (α^*, β^*) minimizing out-of-sample MAD or MSE.
- Plots seasonal decomposition and flags structural changepoints automatically.

Sample Prompts

- "Perform exponential smoothing (α=0.2) and forecast periods 5-8"
- "Detect trend and suggest DES parameters"
- "Compare Naïve vs Moving Average on MAPE"



Workflow 2: Using Gemini for Causal ML & Scenario Stress-Testing

1. Multi-Variable Feature Screening

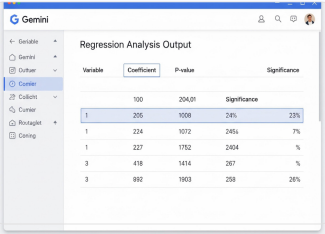
Feed candidate variables (price, promo discount, temperature, age demographics, social sentiment) to run OLS & feature-importance audits (R^2, p-values).

2. Interactive Scenario Stress-Testing

- Real-time “What-If” sensitivity tables (e.g., \$5.50 vs. \$7.00 price $\times$ Pods switch).
- Generates 1,000-trial Monte Carlo scripts to compute optimal Newsvendor production quantities.

Sample Causal Prompts

- "Run linear regression: Demand ~ Price + Ad Spend + Temperature. Interpret coefficients."
- "What happens to demand if we increase ad spend by \$2,000 and temperature rises 5°F?"
- "Identify key market drivers from this dataset and suggest optimal price point."



AI Governance: Benefits, Limitations & Best Practices

Key Benefits

- **Speed & Scale:** Rapidly audits hundreds of SKUs and multi-regional datasets.
- **Scenario Agility:** Immediate calculation of multivariate price/media trade-offs.
- **Accessibility:** Natural language prompts generate reproducible Python/Excel models.

Critical Risks & Limitations

- **Black-Box Risk:** Deep neural nets lack the clear β_i elasticity interpretability of regression.
- **Hallucination Risk:** Never ask an LLM to "guess" a forecast in chat—require code execution & verify MFE/MAD/MAPE.
- **Overfitting & Regime Shifts:** Complex ML can chase historical noise and fail when market rules shift.

Governance Rule: Use AI as an analytical co-pilot alongside transparent baselines (DES, Prophet, Regression) audited by out-of-sample error metrics.

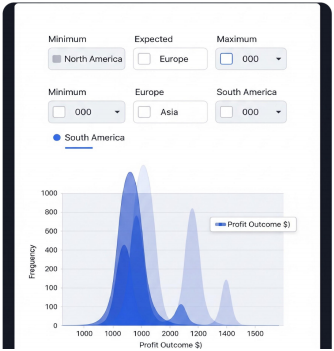
Forecasting in Harvard's Data Analytics Simulation

Managerial Context & Objectives

- You manage **Blue Detergent** for Kelsey-White across 4 US regions (Northeast, South, Midwest, West).
- **Key Decision (Years 5–8):** Set formulation, positioning, channel price, media/trade spend, and **national production quantity**.
- **The Core Challenge:** Maximize cumulative operating profit by solving the **Newsvendor problem** under demand uncertainty.

Two-Stage Decision Architecture

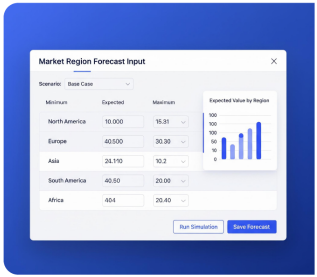
Stage 1: Forecast regional demand distributions.
Stage 2: Convert national demand distribution into a profit-maximizing production order.



Inside the “Forecast Demand” Monte Carlo Tool

How Your Regional Inputs Drive the Simulation

- **1. Four Regional Triangular Distributions:** For each region $r \in \{\text{NE, S, MW, W}\}$, you input $(\text{Low}_r, \text{Expected}_r, \text{High}_r)$.
- **2. 1,000-Trial Monte Carlo Engine:** The tool randomly draws regional demands from your 4 triangles and sums them across 1,000 simulated years.
- **3. National Demand Bell Curve:** By the Central Limit Theorem, summing the 4 regions produces the national mean μ and standard deviation σ used to evaluate your production quantity!



Simulation Forecast Tool

Balancing Forecasts Against Inventory Costs: The Newsvendor Payoff

Asymmetric Economics of "Blue" Detergent

$$\text{Sales} = \min(\text{Demand}, \text{Production} + \text{Inventory})$$

- **Cost of Underage (C_u = Price − Cost):** Huge lost margin ($3–$5/unit) on every stocked-out unit!
- **Cost of Overage (C_o):** Small holding cost, and **unsold inventory rolls over to next year** (not scrapped!).
- **Critical Ratio: CR** $= \frac{C_u}{C_u + C_o} \gg 0.50 \implies Q^* > \mu$.



Concrete Simulation Payoff Rule

Because $C_u \gg C_o$, producing only the mean forecast (33.6M) caps your upside! Buffering toward the upper tail (**48**M units) captures high demand and boosts expected profit by +$**10**M/yr!

Better Forecasting = Higher Simulation Scores

The Profit Engine

- **Unbiased Forecasts:** Captures regional trends and causal elasticity.
- **Cost Efficiency:** Prevents stockouts while controlling rollover inventory.
- **Competitive Advantage:** Maximizes 4-year cumulative profit (Years 5–8).

Top-Tier Student 3-Step Playbook

1. **Descriptive (Audit):** Screen K-W Vision for true signals ($R^2, p < 0.05$).
2. **Predictive (Forecast):** Set (Low, Expected, High) via Trend + Regression ± 2 MAD.
3. **Prescriptive (Order):** Buffer Production + Inventory above mean ($CR > 0.5$).



Takeaway: Descriptive Audit → Predictive Forecast → Prescriptive Newsvendor Buffer = Winning Simulation Strategy!